JAVA OOP LABORATORY EXERCISE

Campus Café Ordering System

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Exercise	Java OOP Laboratory Exercise: Campus Café Ordering System

Introduction

The Campus Café Ordering System is a Java console program that processes customer orders. It supports beverages, pastries, and meals, with a different pricing rule for each type. The program also checks the three-item order limit, applies student discounts or vouchers, and awards loyalty points.

OOP Concepts Used

Concept	How it is used in the program
Abstraction	MenuItem is an abstract class with the abstract computePrice() method. Each menu-item subclass must provide its own price calculation.
Inheritance	Beverage, Pastry, and Meal extend MenuItem and reuse its common fields and methods.
Encapsulation	Private and protected fields protect data. Getters and setters control access and reject invalid values such as negative prices.
Polymorphism	Order stores items as MenuItem references and calls computePrice(). Java runs the correct overridden method for each actual item.
Method Overloading	Customer has addPoints(int) and addPoints(double). Order has computeTotal() and computeTotal(double).
Method Overriding	Beverage, Pastry, and Meal override computePrice() and getDescription() to apply their own rules.
Constructors	Default and parameterized constructors initialize objects. Customer also demonstrates constructor overloading with three constructors.
Conditional Statements	if and if-else statements validate input, calculate prices, enforce the item limit, and apply discounts and vouchers.

Program Output and Explanation

Ana Reyes is a student. Her order contains a large Iced Latte (■120.00), a warmed Ensaymada (■55.00), and an upsized Chicken Adobo with two extra sides (■142.00). The subtotal is ■317.00. After the 10% student discount of ■31.70, her total is ■285.30 and she earns 5 loyalty points. A fourth item is rejected because the order limit is three.

Ben Cruz orders a medium Hot Chocolate (■95.00) and a regular Beef Tapa (■130.00). His subtotal is ■225.00. After the ■20.00 voucher, his total is ■205.00 and he earns 4 loyalty points.

Console Output

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Error: base price cannot be negative.
Order is full. Cannot add Cookie [P-02].

Order #1001 - Ana Reyes (Student)
  Iced Latte [B-01] Size: L                        120.00
  Ensaymada [P-01] (Warmed)                         55.00
  Chicken Adobo [M-01] Upsized, 2 sides            142.00
Subtotal: 317.00
Student discount (10%): -31.70
TOTAL: 285.30
Loyalty points: 5

Order #1002 - Ben Cruz
  Hot Chocolate [B-02] Size: M                      95.00
  Beef Tapa [M-02] Regular, 0 sides                130.00
Subtotal: 225.00
Voucher: -20.00
TOTAL: 205.00
Loyalty points: 4
Error: extra sides cannot be negative.
```

Challenges Encountered and Solutions

One challenge was organizing the program into separate classes and making the objects work together. I used MenuItem as the parent class and stored the customer and up to three MenuItem objects in Order. Another challenge was preventing invalid input and extra items. I used if-else statements in setters and addItem() to reject negative values and a fourth item. I also checked the calculations to make sure the student discount and voucher were applied correctly.

Conclusion

This exercise helped me understand how Java OOP concepts are used in a practical program. I learned to create classes and objects, connect objects, protect data, reuse code through inheritance, and calculate different prices through polymorphism. The program demonstrates the main requirements of the Campus Café Ordering System.